



Valerio Serino

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Date of birth: 29/09/1994

ABOUT ME

Passionate about the full spectrum of computing — from AI and cybersecurity to networks and remote sensing — guided by a green-coding mindset (even when my GPU disagrees) and by a simple principle: *"With great prompts comes great responsibility"*.

PUBLICATIONS

[2021]

[Sensing multi-agent system for anomaly detection on crop fields using phenological context.](#)

Reference: Serino V., et al., 2021. Sensing multi-agent system for anomaly detection on crop fields. ICETCI 2021.

A multi-agent IoT model for crop-field monitoring and anomaly detection.

The system integrates **heterogeneous spectral datasets** — harmonizing **EO** data of different satellite missions (e.g., **Sentinel** and **Landsat** families) — and combines them with **phenological context** to classify vegetation anomalies.

A simulated case study demonstrates the model's ability to promptly detect critical vegetation conditions, burned areas and risk zones following surrounding events.

Journal Name: International Conference on Emerging Techniques in Computational Intelligence (ICETCI), IEEE | **Publisher**: IEEE

WORK EXPERIENCE

Sytel Reply Roma

City: Rome | **Country**: Italy | **Name of unit or department**: Mission-critical enterprise telco systems (clients: TIM and Poste Italiane)

[03/03/2025 – Current]

Senior IT Consultant

- **Designed and optimized high-volume messaging workflows**, implementing complex transformations to core business processes using C, Cassandra, Redis, MongoDB, Oracle DB, SQLite, Filebeat, RabbitMQ and HTTPS/UCP/SMPP based interfaces.
- **Developed and maintained advanced security features**, including CAPTCHA and OTP verification systems, leveraging Oracle Apex, Java and cybersecurity best practices.
- **Researched and engineered offline-capable, queryable knowledge bases** to support AI-ready data workflows, integrating LLMs, LangChain, LangGraph and RAG pipelines.
- **Built a GUI to automate development workflows**, enabling faster prototyping and operational efficiency through Python, REST APIs and LLM-powered components.
- **Coordinated the development team**, performing requirement analysis, technical design, effort estimation and planning of new features and system evolutions.
- **Performed production-level system activities**, including security remediation, server and middleware configuration updates and integrations with internal enterprise systems across PHP/Apache, Docker and VM-based stacks, secured communications, load balancers, reverse proxies and DMZ network segments.

- **Performed analysis, development and vulnerability assessment** on business-critical processes, working with Java, WebLogic, Oracle SQL and IVR systems.
- **Designed and executed remediation actions** on production databases and application servers, ensuring system stability and continuity on HP-UX environments with Apache Reverse Proxy.
- **Led the migration of messaging services** from legacy JMS queues to Apache Kafka, using Docker-based deployments to improve scalability and throughput.
- **Developed data collection workflows and monitoring dashboards** using the ELK stack, Korn shell and Solaris, enabling real-time observability of system performance.
- **Built software adapters for legacy systems**, including vulnerability analysis and secure integration using WebLogic, EJB and MDB components.
- **Implemented security enhancements** on client/server architectures, modernizing components from Java 4 to Java 8 and integrating Java RMI, TLS 1.2, digital certificates and encryption mechanisms.
- **Developed microservices in containerized environments**, leveraging Jenkins, Docker and OpenShift to streamline CI/CD and deployment workflows.

EDUCATION AND TRAINING

[24/09/2018 – 24/03/2020]

Master's degree in Computer Engineering, System and Applications (in English)

The University of Salerno

City: Salerno | **Country:** Italy | **Final grade:** 102 | **Level in EQF:** EQF level 7

Thesis: Intelligent system for monitoring and detecting critical situations on the state of vegetation in forest/agricultural contexts

- **Skills:** Python, Earth Observation, Apache Kafka, Machine Learning, Semantic Web

Main projects:

- Telecommunication Networks - Packet Manipulation and Error Correction (ARQ protocol)
- Cognitive Robotics - Autonomous Robot Soccer (ROS, Gazebo)
- Computer Vision - License Plate Detection and OCR (YOLO + CRNN)
- Cybersecurity - Fake Contact Avoidance in DP3T (replay-attack prevention)

[22/09/2014 – 29/05/2018]

Bachelor's degree in Computer Engineering

The University of Salerno

City: Salerno | **Country:** Italy | **Final grade:** 92 | **Level in EQF:** EQF level 6 | **Thesis:** Design and implementation of a responsive Web App for real-time feedback collection

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING B2 **READING** B2 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Digital Skills:

Earth Observation | Machine Learning | Deep Learning | LLM | RAG (Retrieval-Augmented Generation) | Agent AI | C | Java | Python | Shell Scripting | Web Languages | SQL | Linux | Matlab | Jenkins | WebLogic | Git | Reverse Engineering | Docker | OpenShift | Google Kubernetes Engine (GKE) | n8n

Soft Skills:
Teamwork | Problem Solving | Empathy | Deadline Compliance | Relationship Management

CERTIFICATIONS

- [Cloud Academy, Inc] **Machine Learning on Google Cloud Platform**
- [Cloud Academy, Inc] **Becoming a Cloud Architect - Learn the Fundamentals**
- [Cloud Academy, Inc] **Docker Certified Associate (DCA) Exam Preparation**
- [Cloud Academy, Inc] **Cyber Primer**
- [Cloud Academy, Inc] **Typescript: Zero to Hero**

HOBBIES AND INTERESTS

- Reading about science, engineering, and innovation**
- Robotics & AI experimentation**
- Open-source software and tech communities**

DRIVING LICENCE

Cars: B